



4.Backup and Disaster Recovery

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4.1 Backup strategies for AWS databases

Backups are crucial for **protecting database data** from accidental deletion, corruption, cyberattacks, or system failures. AWS offers multiple backup strategies to ensure data **remains safe and recoverable** at any time.

1. Automated Backups

- AWS automatically takes backups of databases without requiring user intervention.
- Example: Amazon RDS (Relational Database Service) enables daily automatic backups of the entire database instance.
- These backups are stored in Amazon S3 and can be retained for up to 35 days.
- Automated backups include transaction logs, so you can restore data to any point within the backup retention period.
- Helps businesses recover data quickly if something goes wrong, like accidental deletion or corruption.
- Best For: Regular, hands-free backups with scheduled recovery options.

How to create S3 Buckets : [AWS Tutorials - 60 How to Create Bucket in AWS | Create S3 Bucket | Create First B...](#)

2. Manual Snapshots

- Unlike automated backups, manual snapshots are taken manually whenever required.
- These backups are stored indefinitely in Amazon S3 until the user deletes them.
- Useful before making changes to a database (e.g., updating software or making structural modifications).
- Example: Before updating an Amazon RDS database, a snapshot can be taken in case something goes wrong, allowing easy rollback.
- Snapshots can be shared with other AWS accounts, making them useful for testing, development, or migration.

Best For: Taking specific backups before critical updates or system changes.

3. Point-in-Time Recovery (PITR)

- PITR allows restoring a database to any exact time within the backup retention period.
- AWS stores transaction logs, which record all database changes in real-time.
- If a database gets corrupted or important data is mistakenly deleted, it can be restored to a precise point before the issue occurred.
- Example: If a company accidentally deletes some records at 3:00 PM, PITR can restore the database to 2:59 PM, recovering lost data.
- Available for Amazon RDS and DynamoDB.
- Best For: Recovering from accidental data deletion, corruption, or human errors.

4. Cross-Region Backups

- AWS provides an option to store database backups in another AWS region.
- Protects against regional outages, natural disasters, or cyberattacks.
- Example: If a company's primary database is in AWS Mumbai Region, they can store backup copies in the AWS Singapore Region to ensure data is safe even if the primary region fails.
- These backups can be restored to another region, helping in disaster recovery and business continuity.
AWS provides options like Cross-Region Replication (CRR) for some database services like Amazon DynamoDB.
- Best For: Disaster recovery and maintaining data availability across multiple locations.

5. AWS Backup Service

- AWS Backup is a centralized service that automates and manages backups across multiple AWS services.
- It works with Amazon RDS, DynamoDB, Amazon EBS, Amazon S3, and more.
- **Features include:**

- Scheduling backups automatically.
- Setting retention policies (deciding how long backups are stored).
- Monitoring backup status in a single dashboard.
- **Example:** Instead of manually backing up each database separately, a company can use AWS Backup to create and manage backups for all their AWS resources from one place.
- Best For:** Large-scale AWS environments needing automated, policy-driven backups.

6. Multi-AZ Deployments for High Availability

- Not a traditional backup, but a strategy to ensure high availability by replicating the database in multiple AWS Availability Zones (AZs).
- In case of a failure in the primary database, AWS automatically switches to the standby database in another AZ.
- Example: If an Amazon RDS Multi-AZ database in Mumbai (AZ1) fails, AWS automatically switches to a backup copy in Mumbai (AZ2), ensuring zero downtime.
- This minimizes downtime and ensures continuous operations.
- Best For: Business-critical applications that require continuous availability without downtime.

4.2 Automated Backups and Snapshots

Automated Backups

- Automated backups save a copy of your data and system files at scheduled intervals.
- They help in recovering lost data if something goes wrong.
- Cloud providers like AWS, Azure, and Google Cloud offer automated backup services.
- **Retention Policy:** Backups can be stored for a specific period (days, weeks, or months) based on business needs. Older backups may be automatically deleted to save space.

Snapshots

- A snapshot is a point-in-time copy of data (like taking a photo of your system at a specific

moment).

- It captures only the changes made since the last snapshot, saving storage space.
- Snapshots are useful for quickly restoring systems without affecting ongoing operations.
- **Retention Policy:** Snapshots can be retained for short-term recovery or archived for long-term use. Many cloud platforms allow automated snapshot retention policies to manage storage efficiently.

Key Differences Between Backups and Snapshots:

Feature	Backups	Snapshots
Purpose	Full copy of data for recovery	Saves system state at a specific time
Storage	Stored separately from the original data	Stored within the same storage system
Speed	Takes longer, uses more storage	Faster, takes up less space
Retention	Defined period before deletion	Can be short-term or archived
Use Case	Long-term data protection	Quick recovery from failures or changes

Benefits of Automated Backups & Snapshots:

- **Data Protection** – Prevents data loss due to accidental deletion, system failure, or cyberattacks.
- **Fast Recovery** – Snapshots allow quick system rollback if something breaks.
- **Cost-Effective** – Saves storage costs by keeping only necessary copies.
- **Compliance** – Helps businesses follow data protection rules.

4.3 Disaster recovery planning and execution

Disaster Recovery (DR) is a process that helps an organization recover IT systems, data, and applications after unexpected problems like power failures, cyberattacks, or natural disasters. The goal is to minimize downtime and loss of important information.

1. Disaster Recovery Planning

A **Disaster Recovery Plan (DRP)** is a strategy that ensures a company can quickly recover from disruptions. It involves several key steps:

1.1 Risk Assessment

Before making a recovery plan, it is important to identify possible risks that could harm IT systems.

Example:

- A company stores all its customer data on a central server. If the server crashes due to a power failure, all data may be lost.
- A hospital depends on its IT systems to store patient records. If a ransomware attack locks their files, doctors cannot access important medical histories.

Solution: Identify such risks in advance and plan accordingly.

1.2 Critical System Identification

Not all systems in an organization are equally important. Some are **critical** (must work all the time), while others are less essential.

Example:

- A bank's online banking system is critical because customers need 24/7 access.
- A company's internal email system is important but can be restored after a few hours without major problems.

Solution: Prioritize systems and ensure critical ones have faster recovery plans.

1.3 Backup Strategy

Backups are copies of important data stored in safe locations. These backups help restore information if something goes wrong.

Example:

- A school maintains student records on computers. If a system crashes, they can restore data from a backup stored on an external hard drive or cloud storage.
- A company stores project files in Google Drive to avoid losing them due to a local computer failure.

Solution: Use multiple backup methods, including **cloud backups, external hard drives, or offsite storage** to prevent data loss.

1.4 Recovery Objectives (RTO & RPO)

When recovering from a disaster, organizations set two main goals:

- **Recovery Time Objective (RTO):** This is the maximum time a business can afford for a system to be down before it causes big problems. For example, if a company sets an RTO of 2 hours, it means they must fix the issue and restore the system within 2 hours to avoid serious damage.
- **Recovery Point Objective (RPO):** This is the maximum amount of data a business can afford to lose if something goes wrong. For example, if an RPO is 30 minutes, it means the company is okay with losing up to 30 minutes of recent data, but anything more could cause major issues.

Example:

- A bank sets an **RTO of 2 hours** for its ATM network, meaning it must be restored within 2 hours if it crashes.
- An e-commerce site sets an **RPO of 30 minutes**, meaning if the system crashes, they should not lose more than 30 minutes of transaction data.

Solution: Define clear RTO and RPO for each critical system to ensure fast recovery.

1.5 Roles & Responsibilities

In case of a disaster, everyone should know their role. Assigning responsibilities ensures a quick and organized recovery.

Example:

- In a software company, the IT team is responsible for restoring servers, while the security team checks for cyber threats.
- In a hospital, the IT department handles system recovery, while the administration informs doctors about alternative ways to access patient data.

Solution: Create a response team with clearly defined tasks for disaster recovery.

2. Disaster Recovery Execution

When a disaster happens, organizations follow a structured process to restore operations quickly.

Step 1: Assess the Situation

Find out what went wrong and how much damage it caused.

Example: If a company's website is down, the IT team checks if it is due to a cyberattack, server crash,

or power failure.

Step 2: Activate DR Plan

Once the issue is identified, follow the pre-planned disaster recovery steps.

Example: If a company's email server crashes, the IT team follows a step-by-step recovery plan, such as restarting the backup server and restoring email data.

Step 3: Restore Backups

Retrieve lost data from backup storage.

Example: If a university's student database is corrupted, IT staff restores data from the latest cloud backup.

Step 4: Fix and Validate

After restoring systems, check if everything is working properly.

Example: If a company recovers its customer records, employees verify that all files are accessible and no information is missing.

Step 5: Report & Improve

After the disaster is resolved, document the incident and update the DR plan to prevent future issues.

Example: If a bank recovers from a cyberattack, it improves security by adding stronger passwords and two-factor authentication.

3. Types of Disaster Recovery

3.1 Backup-based Disaster Recovery

Storing copies of important data in different locations to restore them if needed.

Example: A school takes weekly backups of student records and stores them on an external hard drive and in Google Drive.

3.2 Cloud-based Disaster Recovery

Using cloud services to store data and recover quickly.

Example: A company using AWS (Amazon Web Services) can restore its website from a cloud backup if the main server crashes.

3.3 Virtualization Disaster Recovery

Running backup systems on virtual machines (VMs) to quickly switch to another system if needed.

Example: If a company's main server fails, a virtual server in another location takes over within minutes, ensuring no downtime for users.

4.4 Best practices for ensuring data durability and availability

Ensuring **data durability** (protection from loss) and **availability** (access when needed) is essential for businesses. Here are five key best practices:

1. **Regular Backups** – Storing multiple copies of data in different locations (e.g., cloud storage, external drives) helps recover lost information in case of failure.
2. **Data Replication** – Keeping copies of data across multiple servers or geographic locations ensures availability even if one system fails.
3. **Cloud Storage with High Uptime SLAs** – Using cloud providers with a **99.99% uptime guarantee** ensures continuous access to data.
4. **Disaster Recovery and Failover Systems** – Having a backup system ready so that if the main system stops working, the backup takes over automatically. This helps keep everything running smoothly without long delays.
5. **Monitoring and Security Measures** – Using **encryption, firewalls, and automated alerts** helps protect data from cyber threats and ensures timely issue detection.

How to Solve Case Study

Example : Solve the following case study with appropriate explanation:

A healthcare organization lost patient data due to an accidental deletion of their AWS-hosted database. How can disaster recovery planning help avoid such losses? What AWS tools and strategies can ensure quick data recovery after accidental deletions?

1. Introduction of Problem (What happened?)

Expected: Explain the problem given in the case study.

Write:

- Which organization is affected
- What problem happened
- What is the impact of that problem

Example: A healthcare organization lost important patient data because their AWS-hosted database was accidentally deleted. This affected hospital operations, patient treatment, and created a risk for sensitive medical records.

2. Cause of Problem (Why did it happen?)

Expected:

Find the main reason behind the issue.

Write:

- Root cause of data loss
- Lack of backup planning
- Weak security or access control

Example: The main reason was accidental deletion of the database and the absence of proper disaster recovery planning. Lack of regular backups, recovery policies, and restricted access made recovery difficult.

3. Solution / Preventive Measures (How to solve?)

Expected: This is the most important part. Write AWS tools and practical solutions.

Write:

- Backup tools
- Recovery methods
- IAM security
- Monitoring services
- High availability strategies

Example:

- Enable AWS RDS automated backups
- Use manual snapshots regularly
- Enable Point-in-Time Recovery (PITR)
- Use AWS Backup for centralized backup management
- Apply IAM least privilege access
- Enable Multi-Factor Authentication (MFA)
- Use Multi-AZ deployment for high availability
- Enable AWS CloudTrail and CloudWatch for monitoring and alerts

4. Conclusion (Final Result)

Expected:

Write the final benefit after applying the solution.

Write:

- Fast recovery
- Reduced data loss
- Better security
- Future prevention

Example: By using disaster recovery planning, automated backups, Point-in-Time Recovery, and strong IAM security, the healthcare organization can quickly recover deleted data and prevent future data loss.

April/ May 2025

Q1 Explain best practices for backup and disaster recovery

Backup and Disaster Recovery (DR) are important to keep data safe and continue work even after system failure or disaster.

1. Follow 3-2-1 Backup Rule: Keep three copies of data, store them on two different media, and keep one copy offsite (like cloud storage such as Amazon S3). This reduces the risk of losing all data at once. If one copy is damaged, other copies are still available. It is a simple and very effective method for data safety.

2. Automate Backups: Use services like AWS Backup to schedule backups automatically. This saves time and reduces human errors. Backups will happen regularly without forgetting. It ensures data is always updated and protected.

3. Define RTO and RPO: RTO means how quickly the system should be recovered after failure. RPO means how much data loss is acceptable. These help in planning backup and recovery strategies. It improves system availability and reduces downtime.

4. Data Security and Encryption: Always encrypt backup data to keep it safe. Use proper access control so only authorized users can access it. This protects data from hackers and unauthorized access. Security is very important for sensitive information.

5. Regular Testing and Monitoring: Test backups regularly to check if data can be restored. Monitor the backup process to detect failures early. This ensures backups are working properly. It avoids problems during real disaster situations.

6. Use Multiple Locations: Store backups in different locations or regions using Amazon Web Services. If one location fails, data is still safe in another place. This protects against natural disasters and system failures. It increases reliability and availability of data.

April/ May 2025

Q2 Describe the AWS backup services and its role in automated database backup

AWS Backup is a fully managed service provided by Amazon Web Services that helps to **centralize and automate data backups** across AWS resources like databases, storage, and file systems.

AWS Backup Services:

AWS Backup allows users to:

- Create **backup plans** to define when and how backups should be taken
- Manage backups of services like Amazon RDS and Amazon DynamoDB
- Store backups securely in backup vaults
- Monitor and manage backups from a single dashboard

It reduces manual effort and ensures consistent data protection.

Role in Automated Database Backup:

1. Automatic Backup Scheduling:

AWS Backup allows setting schedules (daily, weekly). Backups are taken automatically without manual work. This ensures regular and timely database backups.

2. Centralized Management:

All database backups are controlled from one place. No need to manage each database separately. It simplifies administration and saves time.

3. Data Security and Protection:

Backups are encrypted and securely stored. Access is controlled using permissions. This protects sensitive database information.

4. Point-in-Time Recovery:

For databases like Amazon RDS, data can be restored to a specific time. It helps recover from accidental deletion or errors. This improves reliability.

5. Easy Restore and Disaster Recovery:

Data can be restored quickly when needed. Supports disaster recovery planning. Helps maintain business continuity.

